

Midterm Activity – Social Coding Midterm Project

Social Coding Selection

Select a social coding project application for your team from the below options:

- Option 1: Feature enhancements of the Lab 4.9.2 code by adding user-friendly features to the MapQuest REST API [Level of difficulty: +++]
- Option 2: Adapting the Lab 4.9.2 python framework to integrate GPT-3/GPT-4 REST API [Level of difficulty: +++]

What were the reasons your team selected this option?

We selected this option because obtaining a free API key from MapQuest was simpler and more cost-effective compared to integrating the GPT-3/GPT-4 REST API, which would involve usage fees. Additionally, the MapQuest API is similar to the GraphHopper API used in Lab 4.9.2, making it a more straightforward choice for enhancing the existing codebase with user-friendly features.

Describe your team's project application and its deliverables. What are the specific objectives of this application?

Our team's project is focused on developing a route planning and geolocation-based application. The application allows users to input a starting location and a destination to receive directions, along with relevant additional features such as nearby gas stations and hotels. The user can also visualize the route on a map and receive detailed driving directions, which are presented in an easily understandable format. To implement these features, we use the MapQuest API to handle route calculation, geolocation, and the retrieval of nearby services such as gas stations and hotels. The application also allows users to choose their mode of transportation, including car, bicycle, walking, and more, to adapt the route accordingly. For the graphical user interface, we use Tkinter, a Python library that enables us to design a simple, responsive, and user-friendly desktop application. Users can also customize the appearance of the application by selecting from multiple interface themes, including pink mode, dark mode, light mode, and green mode, for a more personalized experience.

Record your team member roles and skillsets

Team member	Role/Knowledge/Skillset
Maël Gouyet	Backend Python developer
Adil Nouri	Front Python developer
Zyad Othman	Backend Python developer
Thibaut Tran	Front Python developer
Angelo Zhou	Front Python developer

Strategy/Project Plan

Provide a brief description of your team's strategy for completing this project.

Our team's strategy for completing this project was centered around clear communication, effective task distribution, and an iterative development process. We communicated primarily through Discord, which allowed us to stay in constant contact, share ideas quickly, and coordinate efficiently. At the beginning of the project, we collectively defined the objectives and identified the key features to implement.

We then divided the work by assigning each team member specific tasks. Using GitHub, we created issues corresponding to these tasks and assigned them accordingly. Each member worked on their own branch and, once a feature was ready, submitted a pull request for review. This iterative approach allowed us to progressively build the application while ensuring code quality and team alignment.

Using GitHub for Collaboration

What is the link to your GitHub repository?

https://github.com/MaelGouyet/software_engineering_team11

Describe how GitHub was used to:

- a. Create branches (in the context of this project)

For each important feature we made an issue and a branch to work on the issue.

- b. Add team members (and their branches/commits)

We added all team members as collaborators on GitHub. Each member worked on their own branch, made commits, and opened pull requests. This allowed us to track individual contributions and merge features into the DEV branch in an organized way. And then we merged DEV into main.

- c. Mention pull requests, code review, merge, etc. (in the context of this project)

When we finished working on the issue we made a pull request to merge our branch into the DEV branch and when we finished important issues we merged DEV into main. When a pull request is made, we ask someone else to review the code and after he reviewed the code we merged the branches.

Final Deliverables

Presentation

Create a presentation about the project you selected. Your presentation should include:

- Information about your application, covering what features your team included
- The reasons that your team decided on these specific features in your application
- Application code including comments and documentation. Your comments and documentation should be sufficient for any other team to be able to continue the project if required. Another
- team should be able to understand the application, your features and how to continue with the project
- Demonstration of the application

- List of future enhancements (backlog)
- Reflection points – what issues have you faced while working on this activity, how did you find solutions, what have you learned, etc.